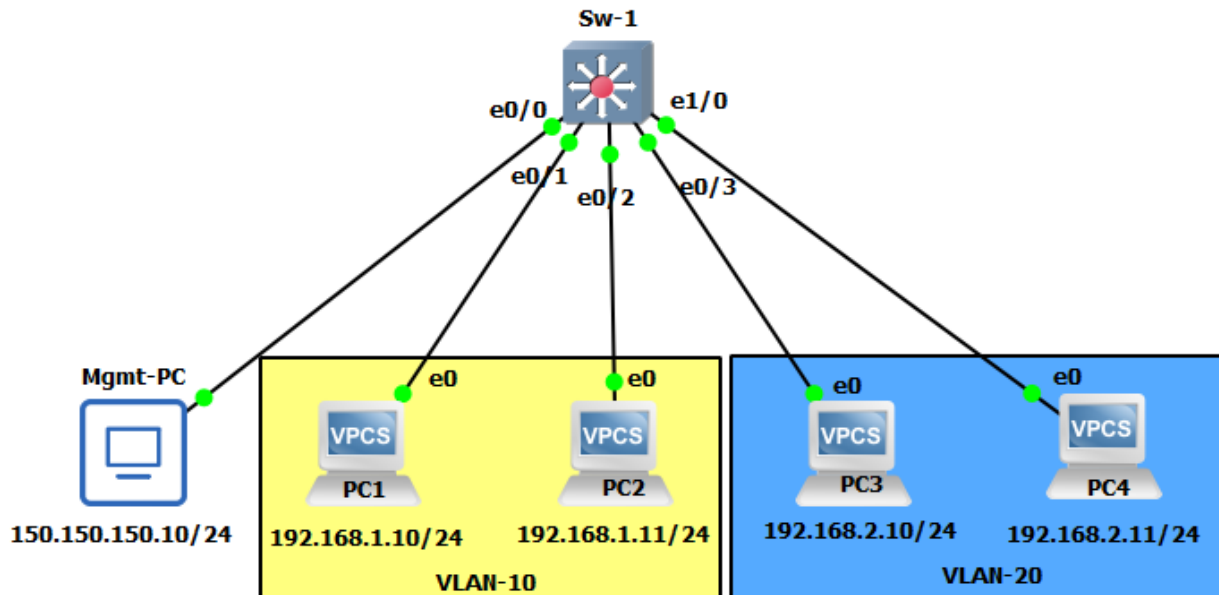


Switch Configuration



Task

1. Configure the switch with IP address 150.150.150.1 for device management and configure enable password as **ccna** and device access password as **cisco**.
2. Configure IP address on VPCs
3. Create VLAN 10 named as Sale and VLAN 20 named as Production.
4. Assign switchports in respective VLAN as per the topology
5. Test the connectivity between VPCs in same & different VLAN.

Note: Mgmt-PC is a cloud which bridge the switch SW-1 with loopback adapter on real PC.

Task-1: Configure the switch with IP address 150.150.150.1 for device management and configure enable password as **ccna** and device access password as **cisco**.

Switch-1 configuration:

```
Sw-1#config t
Sw-1(config)#hostname SW-1
SW-1(config)#no ip domain-lookup
SW-1(config)#enable password ccna
SW-1(config)#int vlan1
SW-1(config-if)#ip address 150.150.150.1 255.255.255.0
SW-1(config-if)#no shut
SW-1(config-if)#exit
SW-1(config)#banner motd $
Enter TEXT message. End with the character '#'.
#####
```

```
#####
#####      Welcome to RSATechforum      #####
#####
#####
$
SW-1(config)#line vty 0 4
SW-1(config-line)#password cisco
SW-1(config-line)#login
SW-1(config-line)#logging synchronous
SW-1(config-line)#exec-timeout 1 0
SW-1(config-line)#exit
SW-1(config)#line console 0
SW-1(config-line)#password cisco1
SW-1(config-line)#login
SW-1(config-line)#logging synchronous
SW-1(config-line)#exec-timeout 0 0
SW-1(config-line)#exit
SW-1(config)#exit
SW-1#
```

Task-2: Configure IP address on VPCs

PC-1 Configuration:

```
PC1> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1
```

PC-2 Configuration:

```
PC2> ip 192.168.1.11/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.11 255.255.255.0 gateway 192.168.1.1
```

PC-3 Configuration:

```
PC3> ip 192.168.2.10/24 192.168.2.1
Checking for duplicate address...
PC1 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1
```

PC-4 Configuration:

```
PC4> ip 192.168.2.11/24 192.168.2.1
Checking for duplicate address...
PC1 : 192.168.2.11 255.255.255.0 gateway 192.168.2.1
```

✓ Verification & Testing:

SW-1#sh vlan brief

VLAN Name	Status	Ports
-----------	--------	-------

```

1 default active Et0/0, Et0/1, Et0/2, Et0/3
Et1/0, Et1/1, Et1/2, Et1/3
Et2/0, Et2/1, Et2/2, Et2/3
Et3/0, Et3/1, Et3/2, Et3/3

1002 fddi-default act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default act/unsup
1005 trnet-default act/unsup
SW-1#

```

Telnet to switch from PC

 Telnet 150.150.150.1

```

#####
##### Welcome to RSATechforum #####
#####
#####
#####

User Access Verification

Password:
SW-1>en
Password:
SW-1#

```

SW-1#sh user

Line	User	Host(s)	Idle	Location
* 0	con 0	idle	00:00:00	
2	vtty 0	idle	00:00:13	150.150.150.10

Interface	User	Mode	Idle	Peer Address

SW-1#sh vlan summary

Number of existing VLANs	: 5
Number of existing VTP VLANs	: 5
Number of existing extended VLANs	: 0

Task-3: Create VLAN 10 named as Sale and VLAN 20 named as Production.

```

SW-1#config t
SW-1(config)#vlan 10
SW-1(config-vlan)#name Sale
SW-1(config-vlan)#exit
SW-1(config)#vlan 20

```

```
SW-1(config-vlan)#name Production
SW-1(config-vlan)#exit
SW-1(config)#exit
SW-1#
```

✓ Verification & Testing:

SW-1#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/0, Et0/1, Et0/2, Et0/3 Et1/0, Et1/1, Et1/2, Et1/3 Et2/0, Et2/1, Et2/2, Et2/3 Et3/0, Et3/1, Et3/2, Et3/3
10 Sale	active	
20 Production	active	
1002 fddi-default	act/unsup	
1003 token-ring-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trnet-default	act/unsup	

SW-1#sh vlan summary

```
Number of existing VLANs      : 7
Number of existing VTP VLANs  : 7
Number of existing extended VLANs : 0
```

Task-4: Assign switchports in respective VLAN as per the topology

```
SW-1#config t
SW-1(config)#int e0/1
SW-1(config-if)#switchport mode access
SW-1(config-if)#switchport access vlan 10
SW-1(config-if)#exit
SW-1(config)#int e0/2
SW-1(config-if)#switchport mode access
SW-1(config-if)#switchport access vlan 10
SW-1(config-if)#exit
SW-1(config)#int range e0/3,e1/0
SW-1(config-if-range)#switchport mode access
SW-1(config-if-range)#switchport access vlan 20
SW-1(config-if-range)#exit
SW-1(config)#exit
SW-1#
```

Verification & Testing

SW-1#sh vlan brief

VLAN Name	Status	Ports
-----------	--------	-------

```

1  default          active    Et0/0, Et1/1, Et1/2, Et1/3
                                Et2/0, Et2/1, Et2/2, Et2/3
                                Et3/0, Et3/1, Et3/2, Et3/3
10  Sale            active    Et0/1, Et0/2
20  Production      active    Et0/3, Et1/0
1002 fddi-default   act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default act/unsup
1005 trnet-default  act/unsup

```

From PC1, ping PC2, PC3 and PC4

PC1> ping 192.168.1.11

```

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=1.460 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=1.239 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.313 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=1.333 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=1.119 ms

```

PC1> ping 192.168.2.10

host (192.168.1.1) not reachable

PC1> ping 192.168.2.11

host (192.168.1.1) not reachable

From PC2, ping PC1, PC3 and PC4

PC2> ping 192.168.1.10

```

84 bytes from 192.168.1.10 icmp_seq=1 ttl=64 time=0.930 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=64 time=1.533 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=64 time=1.340 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=64 time=0.986 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=64 time=1.417 ms

```

PC2> ping 192.168.2.10

host (192.168.1.1) not reachable

PC2> ping 192.168.2.11

host (192.168.1.1) not reachable

From PC3, ping PC4, PC1 and PC2

PC3> ping 192.168.2.11

```

84 bytes from 192.168.2.11 icmp_seq=1 ttl=64 time=0.966 ms
84 bytes from 192.168.2.11 icmp_seq=2 ttl=64 time=1.847 ms
84 bytes from 192.168.2.11 icmp_seq=3 ttl=64 time=1.887 ms
84 bytes from 192.168.2.11 icmp_seq=4 ttl=64 time=1.148 ms
84 bytes from 192.168.2.11 icmp_seq=5 ttl=64 time=1.073 ms

```

PC3> ping 192.168.1.10

host (192.168.2.1) not reachable

PC3> ping 192.168.1.11

host (192.168.2.1) not reachable

PC4> ping 192.168.2.10

84 bytes from 192.168.2.10 icmp_seq=1 ttl=64 time=1.081 ms

84 bytes from 192.168.2.10 icmp_seq=2 ttl=64 time=4.023 ms

84 bytes from 192.168.2.10 icmp_seq=3 ttl=64 time=4.229 ms

84 bytes from 192.168.2.10 icmp_seq=4 ttl=64 time=1.225 ms

84 bytes from 192.168.2.10 icmp_seq=5 ttl=64 time=3.870 ms

PC4> ping 192.168.1.10

host (192.168.2.1) not reachable

PC4> ping 192.168.1.11

host (192.168.2.1) not reachable

To Remove ports and VLANs

SW-1#config t

SW-1(config)#int e0/1

SW-1(config-if)#no switchport access vlan

SW-1(config-if)#exit

SW-1(config)#int e0/2

SW-1(config-if)#no switchport access vlan

SW-1(config-if)#exit

SW-1(config)#int range e0/3,e1/0

SW-1(config-if-range)#no switchport access vlan

SW-1(config-if-range)#exit

SW-1(config)#no vlan 20

SW-1(config)#no vlan 10

SW-1(config)#exit

SW-1#

SW-1#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/0, Et0/1, Et0/2, Et0/3 Et1/0, Et1/1, Et1/2, Et1/3 Et2/0, Et2/1, Et2/2, Et2/3 Et3/0, Et3/1, Et3/2, Et3/3
1002 fddi-default	act/unsup	
1003 token-ring-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trnet-default	act/unsup	
SW-1#		

Important Commands:

Sh vlan brief
Sh vlan summary
Sh user

Ratan